

Thermodynamic State Vector Interface and Second Law Compliance in Biospheric Simulation: Sprint 009 Report

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Sprint 009 Release — <https://github.com/pascalranoroarijaona/WebOfLife>

Abstract

Macro-level biogeochemical cycling (Carbon, Nitrogen, Phosphorus, and Water) within artificial biospheres and planetary Earth pods must fundamentally obey the laws of classical and non-equilibrium thermodynamics. In Sprint 009, we establish the formal thermodynamic state vector interface (`src/thermodynamics/types.ts`) for the *Web of Life* simulation engine. This module bridges macro-scale mass-energy balance with rigorous first and second law constraints. Specifically, we formalize the internal entropy generation rate ($\dot{S}_{\text{gen}}$), the exergy destruction rate ($\dot{I} = T_0 \dot{S}_{\text{gen}}$) via the Gouy-Stodola theorem, and strict boundary flux arrays. To guarantee physical validity across stochastic simulation ticks, state transitions are governed by an immutable `ThermodynamicStateMonad` that intercepts and prevents any violation of the Clausius inequality ($\dot{S}_{\text{gen}} \geq 0$).

1 Introduction and Thermodynamic Foundations

The *Web of Life* simulation models the Earth pod as an open thermodynamic control volume receiving solar irradiance, reradiating longwave thermal radiation to deep space, and maintaining closed-loop material cycles.

1.1 First Law of Thermodynamics (Energy Conservation)

The rate of change of total system internal energy (U_{sys}) is governed by net boundary heat transfers and work interactions:

$$\frac{dU_{\text{sys}}}{dt} = \sum_k \dot{Q}_k - \dot{W}_{\text{net,out}} + \sum_i \dot{m}_i h_i \quad (1)$$

In our closed-material Earth pod, external mass fluxes are zero ($\sum \dot{m}_i = 0$). Energy input is driven exclusively by incoming solar irradiance ($\dot{Q}_{\text{solar}}$) balanced by planetary thermal emission ($\dot{Q}_{\text{emit}}$):

$$\frac{dU_{\text{sys}}}{dt} = \dot{Q}_{\text{solar}} - \dot{Q}_{\text{emit}} + \sum \dot{W}_{\text{in,net}} \quad (2)$$

1.2 Second Law of Thermodynamics (Entropy Balance)

The entropy balance for the open thermodynamic control volume is expressed as:

$$\frac{dS_{\text{sys}}}{dt} = \sum_k \frac{\dot{Q}_k}{T_k} + \dot{S}_{\text{gen}} \quad (3)$$

Isolating the internal entropy generation rate $\dot{S}_{\text{gen}}$ yields the strict non-negativity constraint mandated by the Second Law:

$$\dot{S}_{\text{gen}} = \frac{dS_{\text{sys}}}{dt} - \left(\frac{\dot{Q}_{\text{solar}}}{T_{\text{sun}}} - \frac{\dot{Q}_{\text{emit}}}{T_{\text{sink}}} \right) \geq 0 \quad (4)$$

1.3 Gouy-Stodola Theorem and Exergy Destruction

Thermodynamic imperfection and work potential loss (exergy destruction rate, $\dot{I}$) are linked directly to internal entropy generation via the dead-state temperature (T_0):

$$\dot{I} = T_0 \dot{S}_{\text{gen}} \geq 0 \quad (5)$$

where $T_0 \approx 255$ K represents the effective radiative sink temperature of deep space.

2 System Architecture & Type Contracts

To enforce these physical laws programmatically, we introduce strict TypeScript interfaces in `src/thermodynamics/types.ts`:

```
export interface ThermalBoundaryFlux {
  readonly heatTransferRateWatts: number; // Q_dot (W)
  readonly boundaryTemperatureKelvin: number; // T_b (K)
}

export interface ThermodynamicStateVector {
  readonly timestamp: number;
  readonly systemInternalEnergyJoules: number; // U (J)
  readonly systemEntropyJoulesPerKelvin: number; // S (J/K)
  readonly temperatureKelvin: number; // T (K)
  readonly deadStateTemperatureKelvin: number; // T_0 (K)
  readonly solarInputWatts: number; // Q_solar (W) >= 0
  readonly planetaryEmissionWatts: number; // Q_emit (W) >= 0
  readonly thermalFluxes: readonly ThermalBoundaryFlux[];
  readonly entropyGenerationRateWattsPerKelvin: number; // S_gen_dot (W/K) >= 0
  readonly exergyDestructionRateWatts: number; // I_dot = T_0 * S_gen_dot (W)
  readonly exergyEfficiency: number; // Eta_ex
}
```

3 The Immutable Thermodynamic Monad

State mutations across simulation ticks are wrapped in immutable monad structures (`ThermodynamicStateMonad`) to intercept and reject any physical violations:

```
export class ThermodynamicStateMonad {
  private constructor(private readonly state: ThermodynamicStateVector) {}

  public static initialize(initialState: ThermodynamicStateVector): ThermodynamicStateMonad {
    if (initialState.entropyGenerationRateWattsPerKelvin < 0) {
      throw new Error("Second Law Violation: Initial S_gen_dot < 0.");
    }
    return new ThermodynamicStateMonad(initialState);
  }
}
```

```

}

public map(transitionFn: (current: ThermodynamicStateVector) => ThermodynamicStateVector)
  const nextState = transitionFn(this.state);
  if (nextState.entropyGenerationRateWattsPerKelvin < -1e-9) {
    throw new Error("Second Law Violation: S_gen_dot < 0 detected.");
  }
  return new ThermodynamicStateMonad(nextState);
}

public extract(): ThermodynamicStateVector {
  return this.state;
}
}

```

4 Conclusion

Sprint 009 establishes an unyielding thermodynamic foundation for the *Web of Life* simulation engine. By coupling rigorous First/Second Law formulations with immutable software design patterns, we guarantee that all simulated biogeochemical dynamics remain physically realizable.

Source Repository: <https://github.com/pascalranoroarijaona/WebOfLife>